Circuit Name: The circuit is titled Rock-Paper-Scissors Digital Logic (as designated within the assessment task).

Circuit Function: This circuit will be played by two people in a Rock-Paper-Scissors game made in Logisim software using basic logic gates, such as AND, OR, and NOT gates. The reason for using Logisim is because it lets us build and test digital circuits visually.

For starters, player 1 will use two input switches; S1 and S2. Player 2 will use switches S3 and S4. These switches will be working in twos, for player 1's choice it's S1 and S2 together and for player 2's choice it's S3 and S4.

With each two-bit input a choice is represented = 00 Rock, 01 Scissors, and 10 Paper. 11 is considered the initial state, meaning no actual choice has been made yet, so it is not counted as win or draw.

Three possible outputs are generated after both players submit their inputs, whereafter the circuit will compare these choices and produce a result from the three outputs (P1_Win P2_Win, Draw).

As explained before, the possible cases are Player 1 wins, Player 2 wins, or a Draw. If Player 1 is the winner, then Player 1 wins and outputs 1. If Player 2 wins, the same follows with P2_Win becoming 1. If they both choose the same input as Paper and Paper, then they draw and outputs 1.

The circuit is split into smaller parts called sub-circuits. Player1_Decorder converts Player1's input into either Rock, Paper, Scissors, or the initial state and the Player2_Decorder does the same for Player2. After that, the Win_Logic and Draw_Logic determine the result; breaking it into sub-circuits makes the design cleaner and easier to explain.

Each player's hand signal is represented using two-bit inputs, as stated in the assessment instructions.

Binary choices are as follows: 00 = Rock, 01 = Scissors, 10 = Paper, 11 = Initial State.

Player 1 uses switches S1 and S2, while player 2 uses S3 and S4. The initial state is used before or between rounds and does not count as a win OR a draw.

Circuit Design Explanation: The circuit had sub-circuits because smaller parts made it easier to build and interpret. I split the main circuit into five sub-circuits: Player1_Decoder, Player2_Decoder, P1_WinLogic, P2_WinLogic, and DrawLogic. This guided me step by step before joining the final main circuit.

Initially, Player 1's S1 and S2 inputs are decoded in the Player1_Decoder. It changes the binary input into any four outputs: P_Rock, P_Scissors, P_Paper, and P_Final. Suppose when S1S2 = 00, the output P_Rock becomes 1. For S1S2 = 01, output P_Scissors is 1. When S1S2 = 10, the output P_Paper becomes 1 and when P1P2 = 11, P_Final becomes 1.

Same scenario is applied for Player 2. S3 and S4 inputs of Player 2's go into the Player2_Decoder. It then produces four outputs: Q_Rock, Q_Scissors, Q_Paper, and Q_Final. The letter P represents Player 1's decoded signals, while Q is used for Player 2's. This simplifies reading the circuit and prevents misunderstanding between players.

After the inputs are decoded, the valid signals are sent to the win and draw logic. P1_WinLogic compares the 3 variables where Player 1 wins. Player 1 wins when P_Rock is against Q_Scissors, because Rock beats Scissors or when P_Scissors is against Q_Paper, because Scissors beats Paper. The third Player 1 winning case is when P_Paper is 1 and Q_Rock is 1, because Paper beats Rock.

The P2_Win_Logic works the same way only when Player2 wins. This is checked via AND gates, and OR gates produce the final output.

The Draw_Logic checks if both players pick the same option. A draw happens when both choose the same option (Obviously). The initial state isn't counted as a draw since it is not considered a move, so even if P_Initial and Q_Initial are decoded, they play no part in the win or draw logic.

Everything is connected in the main circuit, the outputs from the decoders are linked to the win and draw logic using labels and tunnels, which helps keep the layout clean and avoids messy wiring. The final outputs are P1_Win, P2_Win, and Draw, clearly showing the result of the game as required in this assignment.

The following expressions are represented with the following symbols, ' and # both mean NOT, . and & mean AND, and + and * mean OR.

P_Rock = S1' . S2' (orS1# & S2#)
P_Scissors = S1' . S2 (or S1# & S2)
P_Paper = S1 . S2' (or S1 & S2#)
P_Inital = S1 . S2 (or S1 & S2)
Q_Rock = S3' . S4' (or S3# & S4#)
Q_Scissors = S3' . S4 (or S3# & S4)
Q_Paper = S3 . S4' (or S3 & S4#)
Q_Initial = S3 . S4 (or S3 & S4)

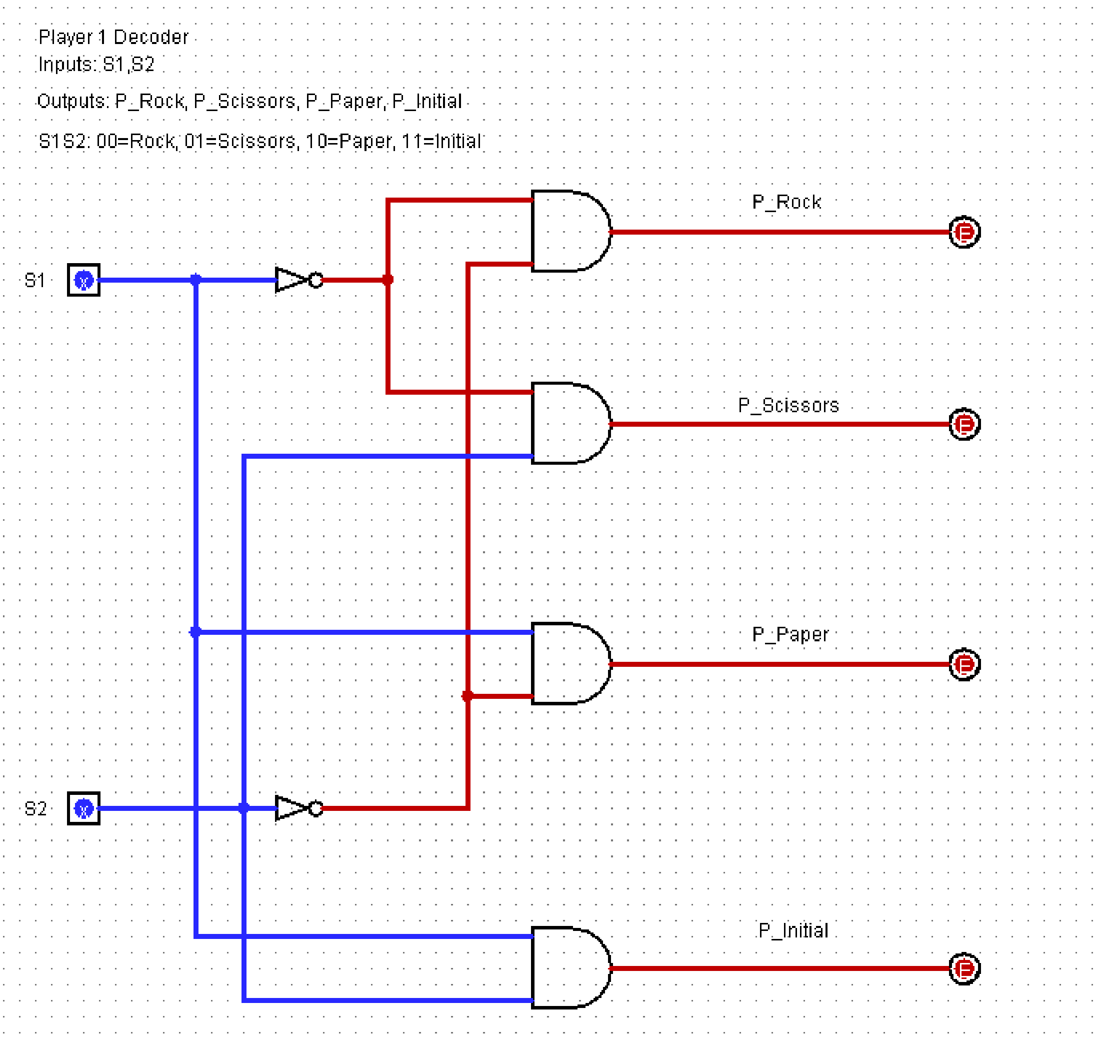
P1_Win is true when Player 1 has a winning combo against Player 2. This can be written as:
P1_Win = (P_Rock . Q_Scissors) + (P_Scissors . Q_Paper) + (P_Paper . Q_Rock)
This basically means Player 1 wins in three situations: Rock beats Scissors, Scissors beats Paper, and Paper beats Rock.

P2_Win is true when Player 2 has a winning combo against Player 1. This can be written as:
P2_Win = (Q_Rock . P_Scissors) + (Q_Scissors . P_Paper) + (Q_Paper . P_Rock)
As before, this means Player 2 wins.

Draw Logic is (P_Rock . Q_Rock) + (P_Scissors . Q_Scissors) + (P_Paper . Q_Paper).
The game is draw when the signals of Player 1 and Player 2 are same.

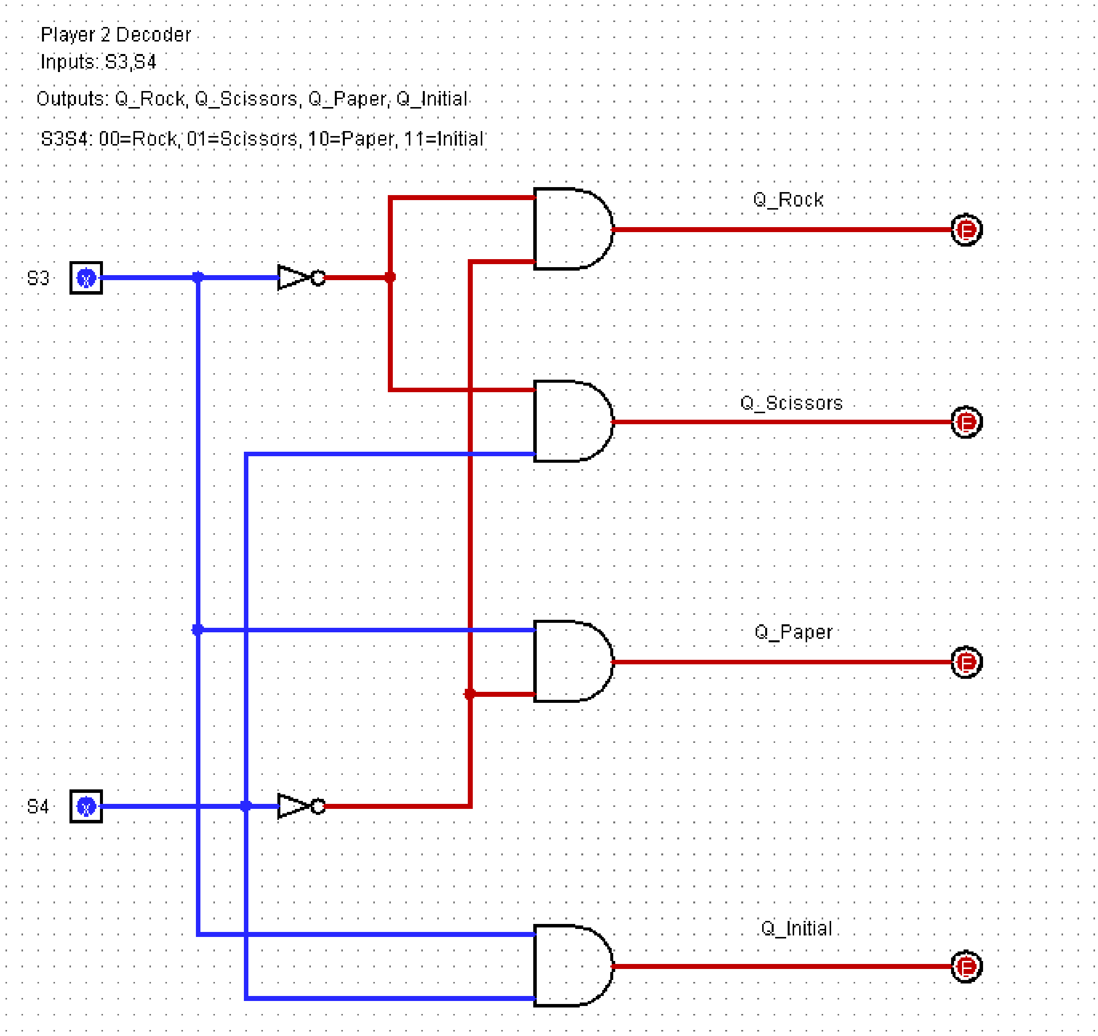
Circuit Diagrams and Screenshots

Figure 1: Player 1 Decoder



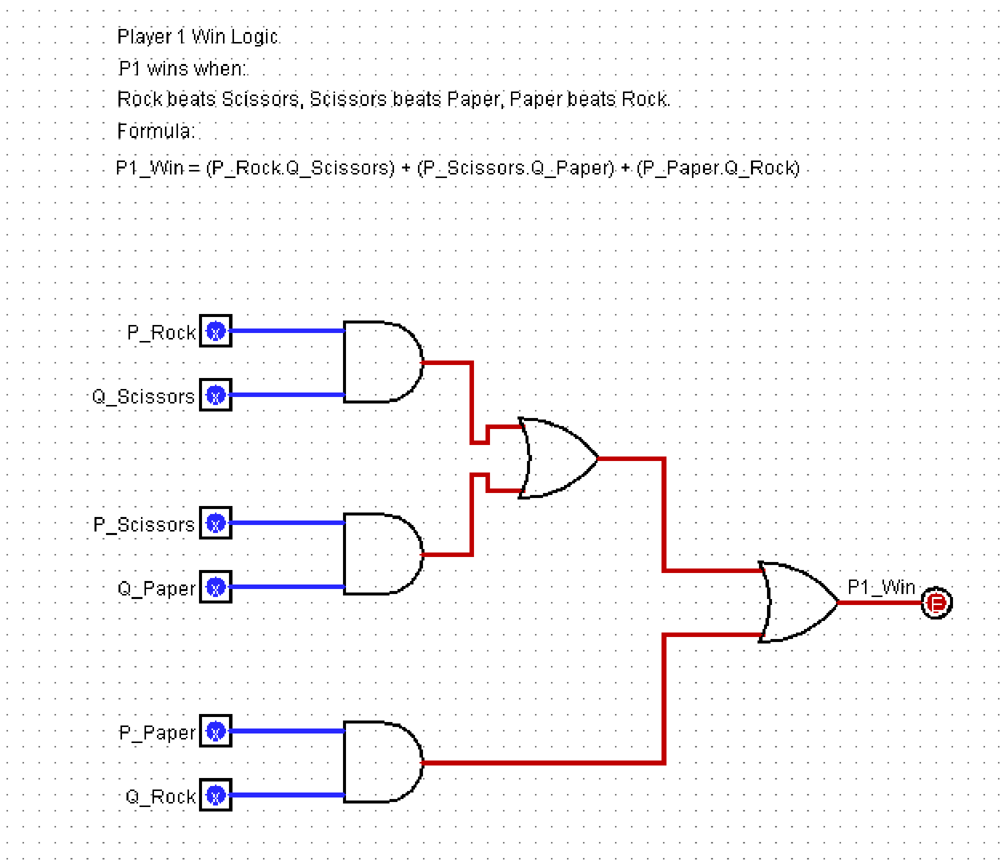
The Player 1 Decoder shows the inputs (S1 and S2) and the outputs (P_Rock, P_Scissors, P_Paper, and P_Initial)

Figure 2: Player 2 Decoder



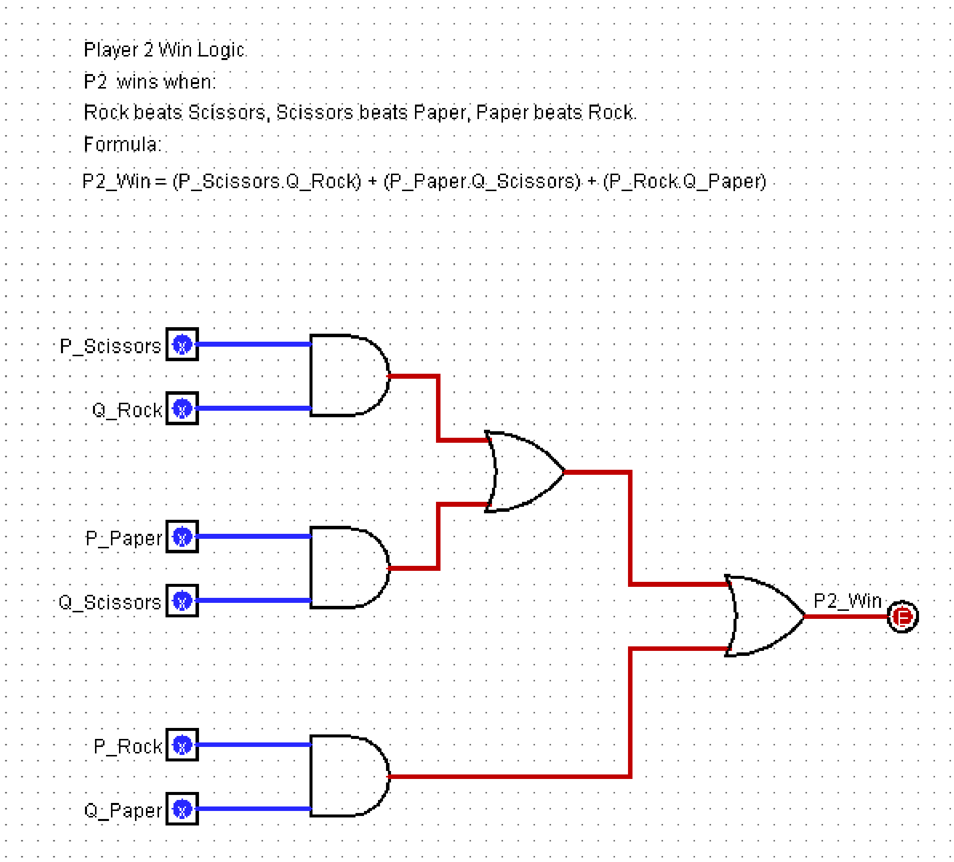
The Player 2 Decoder shows the inputs (S3 and S4) and the outputs (Q_Rock, Q_Scissors, Q_Paper, and Q_Initial)

Figure 3: Player 1 Win Logic



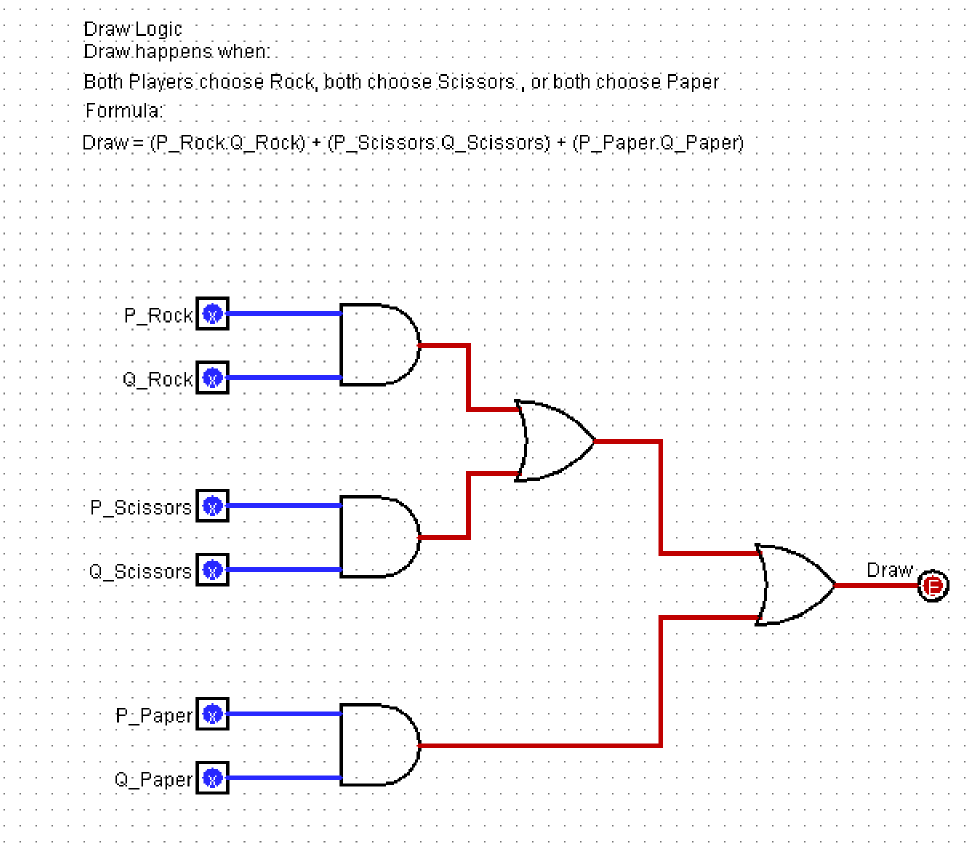
It shows three winning cases for Player 1.

Figure 4: Player 2 Win Logic



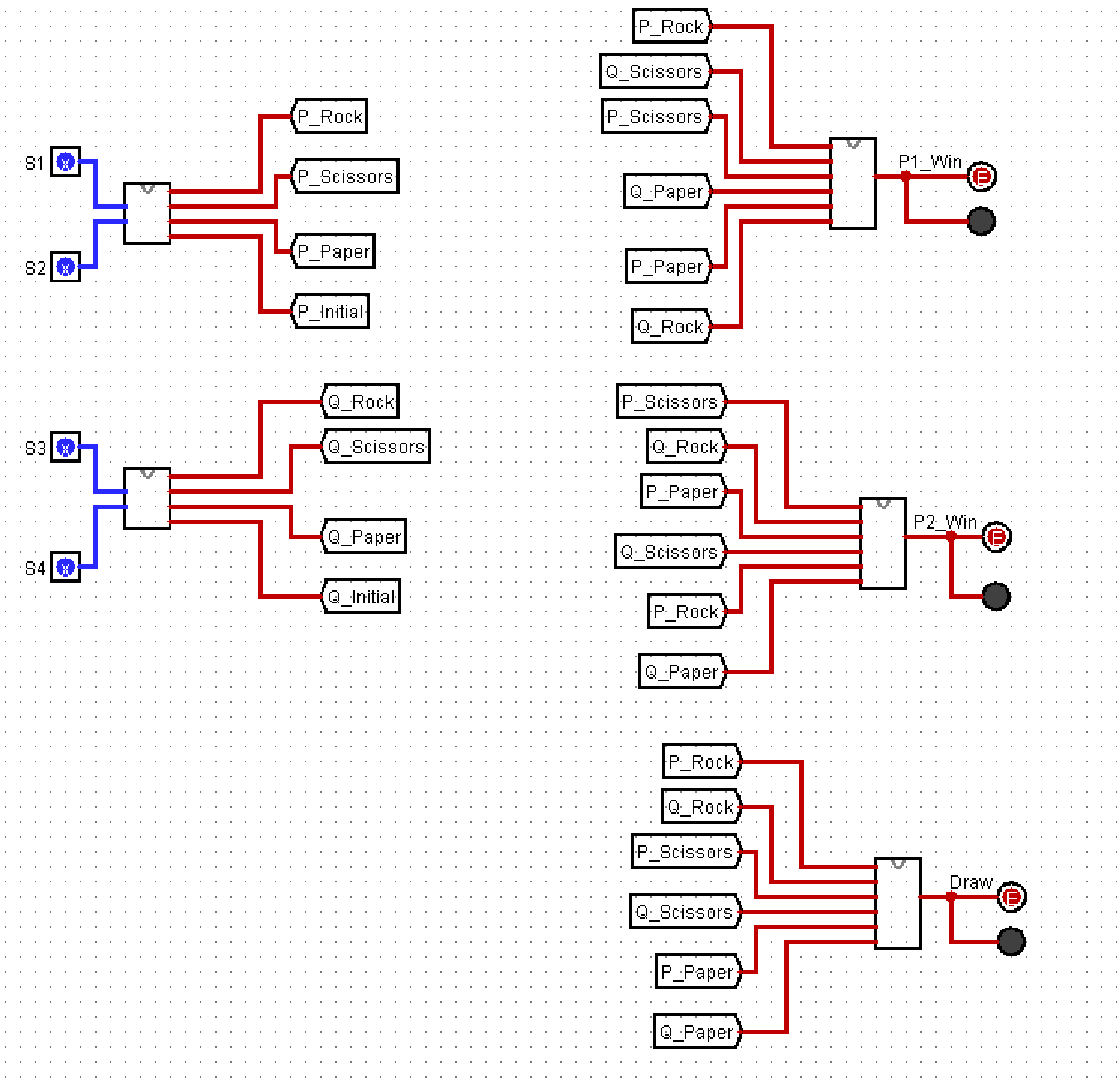
It shows three winning cases for Player 2.

Figure 5: Draw Logic



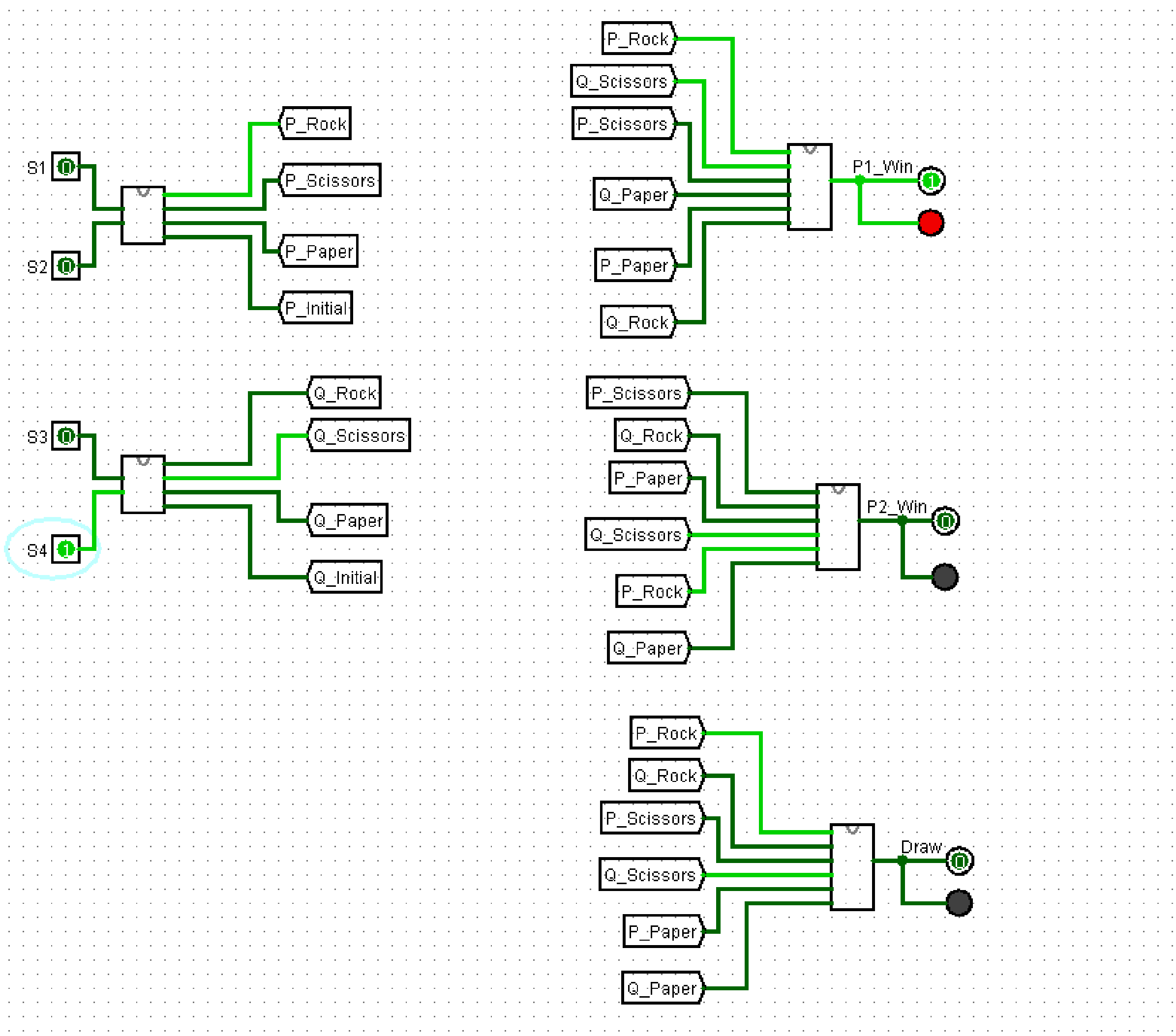
The circuit of Draw Logic indicates the three draw cases.

Figure 6: Main Circuit



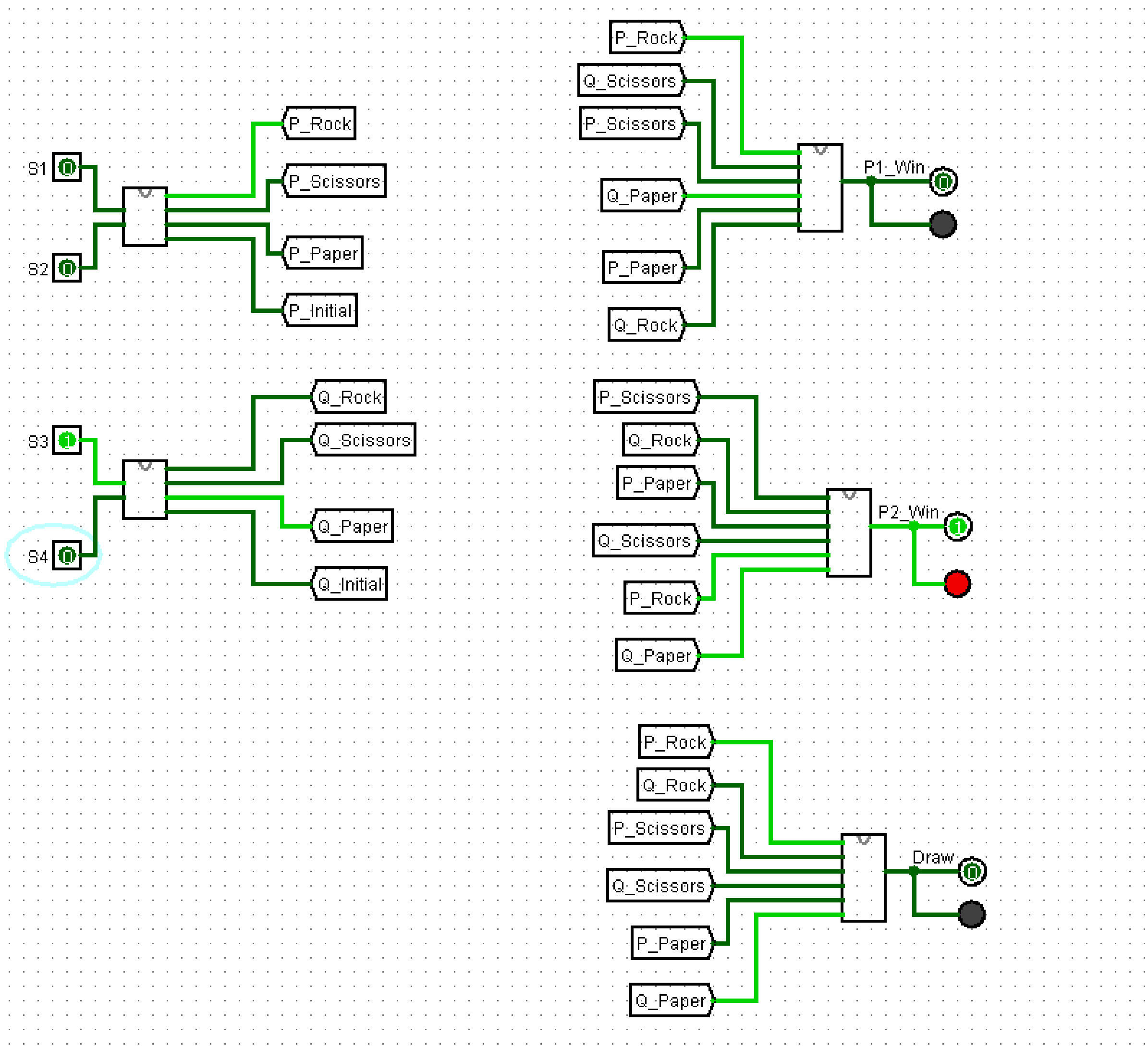
The main circuit connects both decoders and other logic circuits (win logic , draw logic , final outputs)

Figure 7: Player 1 Win Test



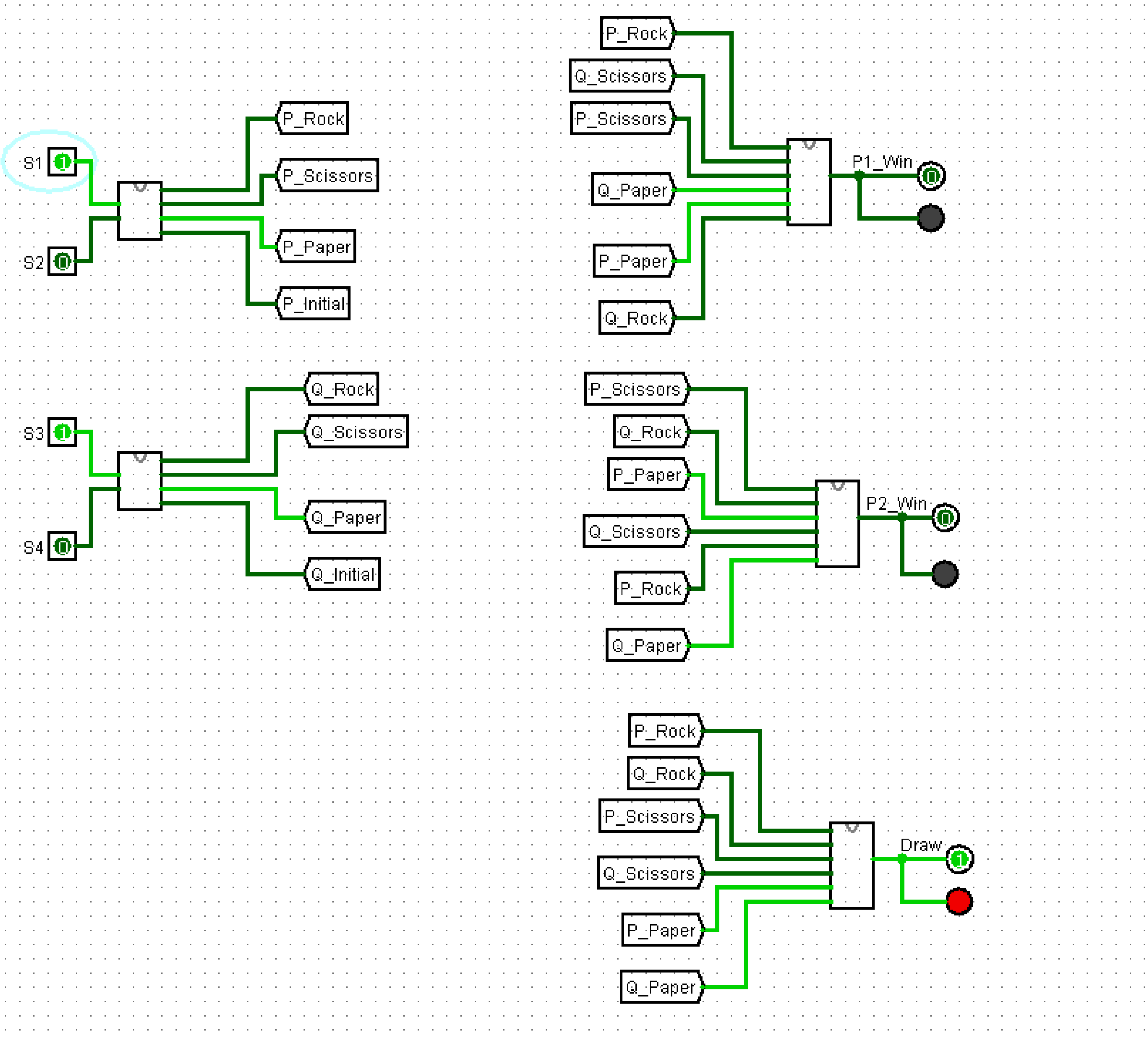
The test case shows the winning scenario of Player 1. As Rock is chosen by Player 1 and Player 2 chooses Scissors.

Figure 8: Player 2 Win Test



The test case shows the winning scenario of Player 2. As Paper is chosen by Player 2 and Rock is chosen by Player 1.

Figure 9: Draw Test



This case shows the draw scenario when both players choose Paper.

Truth Tables

Truth tables are used to show how the circuit behaves for different input combinations. The column headings match the labels in the logisim circuit.

For this circuit, 1 represents the output being active/on, and 0 represents inactivity/ off.

Decoder Truth Table of Player 1:

S1	S2	P_Rock	P_Scissors	P_Paper	P_Initial
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

When S1S2 = 00, Player 1 has chosen Rock. If it’s 01, that is Scissors, and 10 represents Paper. Input 11 representing the initial state.

Player 2 Decoder Truth Table:

Player2_Decoder takes inputs S3 and S4 and converts the two-bit input into Player 2’s chosen signal.

S3	S4	Q_Rock	Q_Scissors	Q_Paper	Q_Initial
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

When S3S4 = 00, Player 2 has chosen Rock. 01 being Scissors, and 10 being Paper. 11 being the Initial State.

Player 1 Win Logic Truth Table

The P1_WinLogic varies the three cases where Player 1 wins that is when Rock beats Scissors, Scissors beats Paper, or Paper beats Rock.

P_Rock	P_Scissors	P_Paper	Q_Rock	Q_Scissors	Q_Paper	P1_Win
1	0	0	1	0	0	0
1	0	0	0	1	0	1
1	0	0	0	0	1	0
0	1	0	1	0	0	0
0	1	0	0	1	0	0
0	1	0	0	0	1	1
0	0	1	1	0	0	1
0	0	1	0	1	0	0
0	0	1	0	0	1	0

The table displays all valid Rock, Paper, and Scissors combos after decoding. Player 1 wins in only three cases as stated before therefore, P1_Win is set to 1 for those rows. For all other combos, P1_Win is 0.

Player 2 Win Logic Truth Table:

The P2_Win_Logic looks at the valid situations where Player 2 wins.

P_Rock	P_Scissors	P_Paper	Q_Rock	Q_Scissors	Q_Paper	P2_Win
1	0	0	1	0	0	0
1	0	0	0	1	0	0
1	0	0	0	0	1	1
0	1	0	1	0	0	1
0	1	0	0	1	0	0
0	1	0	0	0	1	0
0	0	1	1	0	0	0
0	0	1	0	1	0	1
0	0	1	0	0	1	0

This table shows the winning hands after decoding. Player 2 only wins in these three cases: when Player 1 picks Rock and Player 2 picks Paper, when P1 picks Scissors and P2 picks Rock, and when P1 picks Paper and P2 picks Scissors. So, P2_Win is 1 for those rows, and 0 for all other combos.

Draw Logic Truth Table:

The Draw_Logic checks if both players pick the same valid option. A draw happens when both choose Rock, or Scissors, or Paper.

P_Rock	P_Scissors	P_Paper	Q_Rock	Q_Scissors	Q_Paper	Draw
1	0	0	1	0	0	1
1	0	0	0	1	0	0
1	0	0	0	0	1	0
0	1	0	1	0	0	0
0	1	0	0	1	0	1
0	1	0	0	0	1	0
0	0	1	1	0	0	0
0	0	1	0	1	0	0
0	0	1	0	0	1	1

The Table provides information on all the valid Rock, Scissors, and Paper combos after decoding. The output is 1 only when both players choose the same signals. This happens when either Rock against Rock, Paper against Paper, or Scisssors against Scissors. For other combinations. Draw = 0.

The Final Game Output Truth Table shows all the valid possible input combinations for Player 1 and 2 as well as the final output of the game.

S1	S2	S3	S4	Player 1	Player 2	P1_Win	P2_Win	Draw
0	0	0	0	Rock	Rock	0	0	1
0	0	0	1	Rock	Scissors	1	0	0
0	0	1	0	Rock	Paper	0	1	0
0	0	1	1	Rock	Initial	0	0	0
0	1	0	0	Scissors	Rock	0	1	0
0	1	0	1	Scissors	Scissors	0	0	1
0	1	1	0	Scissors	Paper	1	0	0
0	1	1	1	Scissors	Initial	0	0	0
1	0	0	0	Paper	Rock	1	0	0
1	0	0	1	Paper	Scissors	0	1	0
1	0	1	0	Paper	Paper	0	0	1
1	0	1	1	Paper	Initial	0	0	0
1	1	0	0	Initial	Rock	0	0	0
1	1	0	1	Initial	Scissors	0	0	0
1	1	1	0	Initial	Paper	0	0	0
1	1	1	1	Initial	Initial	0	0	0

This table shows that the circuit follows the normal Rock Paper Scissors instructions. The game is draw when the signals are same. When one player's choice beats the other signal, the correct win output is set to 1. If either player is in the initial state, then no win or draw output is triggered.

Testing Results

The final circuit was tested with all different input combinations, and the results show that it follows the Rock Paper Scissors rules as if it were done in-person.

Test case	S1	S2	S3	S4	Meaning	Expected output	Actual outpu
Player 1 wins	0	0	0	1	P1 Rock, P2 Scissors	P1_Win=1, P2_Win=0, Draw=0	Passed
Player 2 wins	0	0	1	0	P1 Rock, P2 Paper	P1_Win=0, P2_Win=1, Draw=0	Passed
Draw	1	0	1	0	P1 Paper, P2 Paper	P1_Win=0, P2_Win=0, Draw=1	Passed
Initial state	1	1	0	0	P1 Initial, P2 Rock	P1_Win=0, P2_Win=0, Draw=0	Passed

In the first test, it shows Player 1 wins When Rock beats Scissors. In the second one, Player 2 wins with Paper against Rock. For the third test, a Draw output is proven when both players choose the same hand signal. The final test proves and shows that the Final state does not create a win or draw output.

Conclusion

The circuit was successfully created following the instructions and tested within logisim. Binary inputs are used for each player, it gives three possible scenarios: Player 1 win, Player 2 win, or a Draw. With the design being split into smaller sub-circuits for decoding, win logic, and draw logic, it made the overall simulation easier to build and understand.

The results of the final testing show the circuit following the rules of Rock-Paper-Scissors accurately. Player 1 wins when Rock beats Scissors, Scissors beat Paper, or Paper beats Rock. Player 2 wins in the opposite cases. Draw output works only when both players choose the same signal. Generally, the circuit meets the required criteria and provides a clear understanding of basic logic gates and design of the circuit.